Description

Image





Caption

1. Golf ball casings are made of ionomer. © Emi Yañez at Flickr - (CC BY 2.0) 2. Golf ball at a golf course © Felix7634 at en.wikipedia - (CC BY-SA 3.0)

The material

Ionomers, introduced by DuPont in 1964, are flexible thermoplastics but they have ionic cross-links, from which they derive their name. Their thermoplastic character allows them to be processed by blow molding, injection molding and thermoforming, and to be applied as coatings. But cooled below 40C they acquire the characteristic of thermosets: high strength, good adhesion and chemical stability.

Compositional summary

Ethylene copolymer containing carboxylic acid groups

General properties

Density		930	-	960	kg/m^3
Price	*	3.82	-	3.96	GBP/kg
Date first used		1965			

Mechanical properties

Young's modulus		0.2	-	0.424	GPa
Shear modulus	*	0.0485	-	0.147	GPa
Bulk modulus		1	-	1.3	GPa
Poisson's ratio	*	0.436	-	0.453	
Yield strength (elastic limit)		8.27	-	15.9	MPa
Tensile strength		17.2	-	37.2	MPa
Compressive strength		9.1	-	17.5	MPa
Elongation		300	-	700	% strain

Hardness - Vickers		2.5	-	4.8	HV
Fatigue strength at 10^7 cycles	*	6.88	-	14.9	MPa
Fracture toughness	*	1.14	-	3.43	MPa.m^0.5
Mechanical loss coefficient (tan delta)	*	0.0943	-	0.286	

Thermal properties

Melting point		80.9	-	95.9	°C
Glass temperature	*	30	-	64	°C
Maximum service temperature		48.9	-	61.9	°C
Minimum service temperature	*	-123	-	-73.2	°C
Thermal conductor or insulator?		Good insulator			
Thermal conductivity		0.239	-	0.276	W/m.°C
Specific heat capacity	*	1.81e3	-	1.89e3	J/kg.°C
Thermal expansion coefficient		180	-	306	µstrain/°C

Electrical properties

Electrical conductor or insulator?		Good insulator			
Electrical resistivity		3.3e21	-	3e22	µohm.cm
Dielectric constant (relative permittivity)		2.2	-	2.4	
Dissipation factor (dielectric loss tangent)	*	0.00295	-	0.00305	
Dielectric strength (dielectric breakdown)		15.7	-	17.7	MV/m

Optical properties

Transparency		Transparent			
Refractive index		1.5	-	1.52	

Critical Materials Risk

High critical material risk?		No			

Processability

Castability		4	-	5	
Moldability		4	-	5	
Machinability		3	-	4	
Weldability		5			

Durability: water and aqueous solutions

Water (fresh)		Excellent			
Water (salt)		Excellent			
Soils, acidic (peat)		Excellent			
Soils, alkaline (clay)		Excellent			

Values marked * are estimates.
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Wine	Limited use

Durability: acids

Acetic acid (10%)	Excellent
Acetic acid (glacial)	Unacceptable
Citric acid (10%)	Excellent
Hydrochloric acid (10%)	Excellent
Hydrochloric acid (36%)	Excellent
Hydrofluoric acid (40%)	Excellent
Nitric acid (10%)	Excellent
Nitric acid (70%)	Unacceptable
Phosphoric acid (10%)	Excellent
Phosphoric acid (85%)	Excellent
Sulfuric acid (10%)	Excellent
Sulfuric acid (70%)	Unacceptable

Durability: alkalis

Sodium hydroxide (10%)	Excellent
Sodium hydroxide (60%)	Excellent

Durability: fuels, oils and solvents

Amyl acetate	Unacceptable
Benzene	Limited use
Carbon tetrachloride	Unacceptable
Chloroform	Unacceptable
Crude oil	Limited use
Diesel oil	Excellent
Lubricating oil	Limited use
Paraffin oil (kerosene)	Limited use
Petrol (gasoline)	Limited use
Silicone fluids	Acceptable
Toluene	Unacceptable
Turpentine	Limited use
Vegetable oils (general)	Acceptable
White spirit	Limited use

Durability: alcohols, aldehydes, ketones

Acetaldehyde	Limited use
Acetone	Excellent

Ethyl alcohol (ethanol)	Limited use
Ethylene glycol	Limited use
Formaldehyde (40%)	Limited use
Glycerol	Excellent
Methyl alcohol (methanol)	Limited use

Durability: halogens and gases

Chlorine gas (dry)	Unacceptable
Fluorine (gas)	Unacceptable
O2 (oxygen gas)	Unacceptable
Sulfur dioxide (gas)	Excellent

Durability: built environments

Industrial atmosphere	Excellent
Rural atmosphere	Excellent
Marine atmosphere	Excellent
UV radiation (sunlight)	Fair

Durability: flammability

Flammability	Highly flammable

Durability: thermal environments

Tolerance to cryogenic temperatures	Unacceptable
Tolerance up to 150 C (302 F)	Acceptable
Tolerance up to 250 C (482 F)	Unacceptable
Tolerance up to 450 C (842 F)	Unacceptable
Tolerance up to 850 C (1562 F)	Unacceptable
Tolerance above 850 C (1562 F)	Unacceptable

Primary material production: energy, climate change and water

Climate change (CO2-eq) (virgin grade production)	* 5.23	-	5.77	kg/kg
Embodied energy (virgin grade production)	* 107	-	118	MJ/kg
Water usage	* 267	-	295	l/kg

Material processing: energy

Polymer extrusion, embodied energy	* 8	-	8.89	MJ/kg
Polymer molding (injection, blow), embodied energy	* 19.2	-	23.3	MJ/kg
Machining, coarse, embodied energy	* 15.5	-	17.2	MJ/kg
Machining, fine, embodied energy	* 17.3	-	19.5	MJ/kg
Machining, grinding, embodied energy	* 19.1	-	21.7	MJ/kg

Material processing: climate change

Polymer extrusion, climate change (CO2-eq)	* 0.525	-	0.586	kg/kg
Polymer molding (injection, blow), climate change (CO2-eq)	* 1.31	-	1.62	kg/kg
Machining, coarse, climate change (CO2-eq)	* 1.01	-	1.13	kg/kg
Machining, fine, climate change (CO2-eq)	* 1.13	-	1.28	kg/kg
Machining, grinding, climate change (CO2-eq)	* 1.25	-	1.42	kg/kg

Recycling and end of life

Recycle	✓			
Functional recycle	✗			
Climate change (CO2-eq), recycling	* 1.73	-	1.9	kg/kg
Embodied energy, recycling	* 20.3	-	22.4	MJ/kg
Recycle fraction in current supply	1			%
Combust for energy recovery	✓			
Heat of combustion (net)	* 36.4	-	38.3	MJ/kg
Combustion CO2	* 2.68	-	2.82	kg/kg
Landfill	✓			
Biodegrade	✗			
Toxicity rating	Non-toxic			
A renewable resource?	✗			

Environmental notes

Ionomers have properties that resemble thermosets, yet they can be recycled like thermoplastics -- an attractive combination.

Recycle mark



Supporting information

Design guidelines

Ionomers are very tough, they have high tensile strength and excellent impact, tear, grease and abrasion resistance. Optical clarity is also quite high. They are most often produced as film. Ionomers have outstanding hot tack (10 times that of LDPE). Their resistance to weather is similar to that of PE, and like PE, they can be stabilized with the addition of carbon black. Permeability is also similar to that of PE, except for carbon dioxide where the permeability is lower. Low temperature flexibility is excellent but they should not be used at temperatures above 71 C. Because of the ionic nature of the molecules, ionomers have good adhesion to metal foil, nylon and other packaging films. Foil and extrusion coating are common. Ionomers have higher moisture vapor permeability (due to the low crystallinity) than polyethylene, are easily sealed by heat and retain their resilience over a wide temperature range.

Technical notes

Ionomers are co-polymers of ethylene and methacrylic acid. Some grades contain sodium and those have better optical properties and grease resistance; some contain zinc and have better adhesion. The ionic crosslinks are stable at room temperature, but break down upon heating above about 40 C. The advantages of crosslinking are seen in the room temperature toughness and stiffness. At high temperatures the advantages of linear thermoplastics appear - ease of processing and recyclability.

Typical uses

Food packaging, athletic soles with metal inserts, ski boots, ice skate shells, wrestling mats, thermal pipe insulation, license plate holders, golf ball covers, automotive bumpers, snack food packaging, blister packs, bottles.

Tradenames

Bexloy, Formion, Iotek, Lucalen, Surlyn

Links

ProcessUniverse

Producers

Reference

Values marked * are estimates.
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